

SOHAM MAHAJAN

(508)-509-5707 | smahajan1101@gmail.com | linkedin.com/sohammahajan | github.com/sohammahajan | Boston, MA

SKILLS

- **Languages / Frameworks:** Python, C/C++, Go, SQL, R, Bash, Flask, FastAPI, PySpark
- **AI / ML:** PyTorch, TensorFlow, Hugging Face, LangChain, BERT, RAG, MLflow, Keras, CUDA, SpaCy
- **Database / Vector Stores:** Oracle, MySQL, PostgreSQL, MongoDB, DynamoDB, Snowflake, Teradata, Pinecone, Qdrant, FAISS
- **Cloud & MLOps:** AWS (S3, Glue, SageMaker, EC2, Redshift), Azure, Docker, Kubernetes, GitHub Actions, Jenkins

EDUCATION

University Of Massachusetts

Jan 2024 – Aug 2025

Master of Science - Computer Science | GPA: 3.85/4

- Vice President and College of Engineering Senator, Graduate Student Senate (GSS)
- **Relevant Coursework:** Advanced Machine Learning, Artificial Intelligence, Big Data Analytics, Database Design

University Of Mumbai, India

June 2019 - May 2023

Bachelor of Engineering - Computer Engineering | CGPA: 8.93/10

EXPERIENCE

Bajaj Allianz

Jan 2023 – Dec 2023

Machine Learning Intern

- Designed and implemented real-time data pipelines using **AWS** and **Databricks**, improving processing efficiency time by 20%.
- Developed ML models for predictive inventory management using **time-series forecasting** and **anomaly detection** algorithms to anticipate stock demand, preventing shortages.
- Developed **AI chatbot** for inventory tracking, reducing stockout incidents by 20% through ML-based purchase predictions.
- Created real-time data pipelines integrated with **Tableau**, enabling executive dashboards with predictive insights, cutting decision lag by 35%.

DIGITebl Private Limited

Sep 2020 – Jan 2021

Software Engineer Intern

- Built **REST APIs** in Python using **Flask** and **PyTest**, improving backend performance.
- Automated deployments with **Bash scripting** and **GitHub Actions**, enhancing system stability with integration testing.
- Collaborated in **Agile sprints**, refactored the legacy codebase and improved system throughput by 15%.

RESEARCH EXPERIENCE

Adaptive Weight Decay and Batch Normalization for Sparse Feature Learning

Feb 2025 - Present

- Researched how adaptive weight decay methods like **AdaDecay** and **adaptive learning rate** schedules affect feature sparsity, model convergence and generalization in neural networks.
- Analyzed models under **Gaussian** and **structured noise** to evaluate robustness and gradient behavior.
- Compared performance of **AdaDecay** and **AdamW** optimizers, focusing on training stability, sparsification and efficiency.
- Conducted extensive experiments using **PyTorch**, **NumPy**, **Matplotlib**, step size variations and decay factors to enhance network performance.

TECHNICAL PROJECTS

AI-Enabled Natural Language Interface for Redshift

- Built a natural language interface using the **OpenAI API** to translate English queries into **SQL** for **Redshift**, improving accessibility and enabling seamless integration with **Power BI** for real-time visualization.
- Implemented contextual suggestions, anomaly detection and predictive analytics to enhance decision-making.

DocuQuery: Contextual Document Retrieval and QA System

- Engineered scalable pipelines to process 200K+ PDF files using **transformer-based** models, sentence embeddings and **vector databases (FAISS)**.
- Enabled document-based question answering using **OpenAI API**, **LangChain** and **Retrieval-Augmented Generation**, with integrated **OCR** and **semantic search**.
- Deployed across domains like finance, operations and marketing for insights.

AutoDeployX: CI/CD Automation Framework

- Designed a full **CI/CD pipeline** that runs **unit tests**, performs linting, builds **Docker** images and scans for vulnerabilities.
- Integrated **GitHub Actions** for automatic deployment to an **EC2**-hosted environment with post-deployment alerts sent to **Slack**.